Technical Proposal

Unified Gateway Solution

for

Jindal Steel Odisha Limited,
Angul (JSOL)

(Ref No - 25/JSOLA/UGS/01/v1)

17th Nov. 2025 Ver. 1

Hitachi India Pvt Ltd.

www.hitachi.co.in | ***sales.paeg@hitachi.co.in***

Sushil Signed by Sushil , Date: 20-Nov-2025 05:46 PM +05:30, DocID: cb5c2900-e9e2-4def-9100-51ad7633c5d9.

S. Randive Signed by Swapnil Randive, Date: 20-Nov-2025 05:48 PM +05:30, DocID: cb5c2900-e9e2-4def-9100-51ad7633c5d9.

Mahesh Patil Signed by Mahesh Patil, Date: 21-Nov-2025 09:50 AM +05:30, DocID: cb5c2900-e9e2-4def-9100-51ad7633c5d9.

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ABBREVIATIONS

UGS: Unified Gateway Solution
HIL: Hitachi India Pvt. Ltd.
BF: Blast Furnace
SIN: Sinter
RMHS: Raw Material Handling System
SV: Supervisor
HMI: Human Machine Interface
PLC: Programmable Logic Controller
L1: Level-1 system
L2: Level-2 system
L3: Level-3 system

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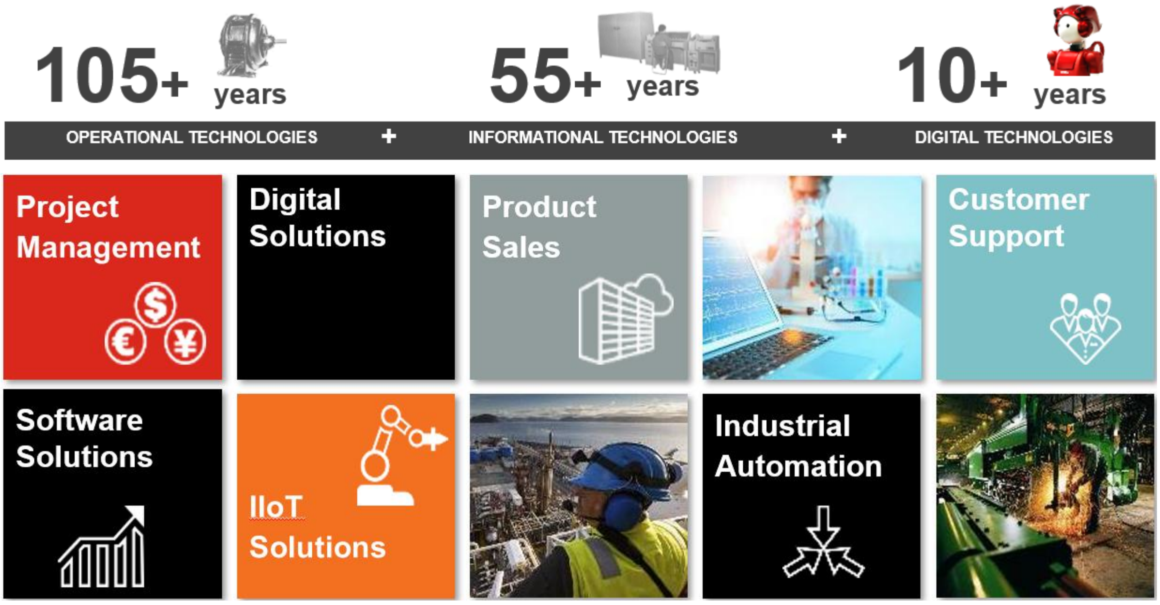
1. EXECUTIVE SUMMARY

Hitachi India Pvt. Ltd. (hereafter also referred to as HIL) provides automation solutions and digital solutions designed and developed for the industrial sector, particularly the metals industry. It takes advantage of more than 105 years of OT (Operation Technology) experience with process knowledge experts & SMEs and 55+ years of IT experience with the newest automation technology.



HIL is uniquely qualified to provide the control system and digital solutions you desire due to our in-house team of well-trained and certified professionals with the right experience & knowledge in designing, developing, and deploying the best solutions possible over a wide range of sectors & areas.

Some of our clients include:

2. GENERAL OVERVIEW

This document is a technical proposal against your ***enquiry shared during online discussion and email dated-23/10/2025*** *for implementing Unified Gateway Solution at various unit facilities of M/s Jindal Steel Odisha Limited (JSOL), Angul, Odisha, India.*

A single standalone IT solution cannot solve the complex needs of a manufacturing setup. But the overall solution stack should be tightly integrated to bring synergies into the manufacturing and business processes.

HIL is offering a solution that will meet M/s JSOL, Angul requirements and will bring significant value to your business. This proposal offers a brief overview into the system modules that can be of significance to different production plants like Blast Furnace, Sinter and RMHS in Jindal Steel Odisha Limited, Angul.

Using latest proven technologies, controllers and tools HIL can implement a comprehensive digitalization framework system by integrating with existing L2 automation system, L1 automation system, energy-meter SCADA system & other interfacing equipment/systems.

The key to the benefits of **Digitalization** lies in balancing and measuring digital investments across strategic focus areas such as, including primary processes like Blast Furnace and RMHS day-to-day operations, digital system reliability, plant productivity and future scalability.

Each focus area is associated with specific metrics and related key performance indicators (KPIs) that allow companies to track and measure the impact of their digital initiatives more accurately.

It also sets the stage for a phased Digitalization, IT & business process maturity program, which aims at bringing alignment across the business & manufacturing processes that help in faster realization of organizational goals.

3. PLANT TECHNICAL DATA

3.1 PRODUCTION UNIT OVERVIEW

The existing production units & equipment to be considered in the solution are as follows:

1. Blast Furnace
2. Sinter
3. RMHS

3.2 PLANT TECHNICAL DATA

Present automation level as communicated by buyer in Request for Proposal (RFP) is shown below:

S. No	Production Unit	Existing HMI & PLC Make & Model	HMI & PLC Software Version	Comm. Interface
1	BF	IBA PDA Server	Version 8	OPC UA
2	SIN	IBA PDA Server	Version 8	OPC UA
3	RMHS	PLC: ABB	PM866, AC 800M	OPC UA / TCP IP
		PLC: Siemens	CPU412-5H (6E7412-5HK06-0AB0)	
		IBA PDA Server	Version 8	

Note: Data Format, Telegram or Data Structure of interface with existing IBA PDA server or PLC's with UGS system to be decided along with Buyer during detail engineering phase.

4. FUNCTIONAL SPECIFICATIONS

4.1 INFORMATION

The **Unified Gateway Solution (UGS)** is an advanced industrial data integration platform designed to centralize, standardize, and securely transfer process data from various steel plant units — including **Blast Furnace, Sinter Plant and RMHS** — into a unified digital framework.

UGS shall communicate directly with **IBA PDA Servers**, collecting real-time operational data and ensuring seamless connectivity across multiple control systems and equipment. The system eliminates the need for multiple vendor tools by providing a single, unified interface for data acquisition and communication management.

This system is planned to be installed as a framework for centralized data storage consisting of 20,000 tags data point & 1 year retention policy in the on-premises UGS server for the production facilities at JSOL, Angul. The system will also act as OPC-UA server to enable the data transfer over OPC-UA protocol to the existing FDTM Client system.

As part of this package, the **HX-Hitachi Controller Device** is supplied and integrated with the Unified Gateway. This industrial-grade controller acts as an edge device to ensure real-time reliable data collection, data buffering, preprocessing, and secure data forwarding to the central UGS application — even under intermittent network conditions

The proposed solution acts as a robust **data acquisition and communication platform**, ensuring that process parameters, equipment statuses, and key operational indicators from all production units are available centrally for analytics, reporting, and future integration with higher-level systems such as MES, ERP, or Cloud Platforms. Unified Gateway Solution (UGS) forms a critical layer for plant digitalization, enabling robust data flow, operational traceability, and system transparency across production units.

The offered UGS system includes:

a) Centralize Data Acquisition
b) User-Friendly Configuration
c) Integration with HX-Hitachi Controller
d) Real-Time Log Monitoring and Diagnostics
e) User Operation Tracking
f) Multiple Protocol Support
g) Advance House Keeping Feature for Space utilization
h) Email alert and reporting system
i) Remote support for six (6) months
j) User Accounts
k) Documentation Control
l) Customer Training

4.2 UNIFIED GATEWAY SOLUTION FEATURES

4.2.1 Centralized data Acquisition:

1

The **Unified Gateway Solution (UGS)** establishes a robust and centralized data acquisition framework for real-time collection of process data from **Blast Furnace, Sinter Plant and RMHS**. Each plant unit's data is fetched directly from the respective **IBA PDA Servers** using the **OPC UA protocol** at the polling frequency of 10 sec, ensuring standardized, secure, and high-speed data communication. The system continuously acquires critical process parameters as per operation requirements and events with configurable data frequency and communication intervals. It also allows users to collect data from existing Level-2 servers of Blast Furnace and Sinter at a minimum polling frequency of 1 minute based on specific operational needs. This overall allows optimization of data acquisition performance based on process requirements.

In addition to real-time communication, the Unified Gateway also acts as Data Concentrator system, which enables historical data tracking and supports long-term storage through integration with **Centralized databases**. The data can be extracted from the Data Concentrator by any external on-premises system with standard protocol connection or SQL query. However, to preserve availability and responsiveness it is recommended to avoid heavy SQL queries which may impact on data concentrator performance.

4.2.2 User Friendly Configuration:

The **Unified Gateway Solution (UGS)** offers an intuitive, user-friendly web-based configuration interface designed to simplify system setup and reduce engineering time. All configurations — including protocol setup, tag mapping, and data flow registration — can be completed in just a few steps through a clean web-based UI screen. Users can easily define source and destination parameters, set communication frequency, and manage protocol-specific options without requiring any programming expertise. The configuration process has been designed for speed, simplicity, and accuracy, ensuring quick deployment even across multiple plant units.

Through this centralized interface, engineers can monitor communication status, modify configurations on the fly, and verify data flow integrity. Each change or update is immediately reflected across the system, ensuring real-time adaptability. By eliminating complex manual configurations and standardizing the setup process, the UGS drastically minimizes commissioning time, reduces configuration errors, and enhances overall operational efficiency within the steel plant's data communication framework.

3-Step Configuration Process covering:

1. **Protocol Configuration** – Select and configure industrial or IoT protocols.
2. **Tag Mapping** – Define and map process parameters or data points.
3. **Flow Registration** – Register and manage inbound/outbound data communication routes.

4.2.3 Integration with HX-Hitachi Controller:

The **Unified Gateway Solution (UGS)** is seamlessly integrated with the **HX-Hitachi Controller Device**, which serves as an intelligent edge component within the overall data acquisition architecture. The HX controller is responsible for real-time data collection, local buffering, and preprocessing before forwarding information to the Unified Gateway. This design ensures high-reliability data acquisition and prevents data loss during network disruptions or system downtime. Acting as a bridge between plant-level control systems and the UGS, the HX controller enhances stability and communication efficiency in continuous industrial operations.

In addition to its hardware reliability, the HX-Hitachi Controller supports **high-level programming languages such as C and C++**, enabling flexible and efficient data collection logic. It is also compatible with the **open-source CODESYS platform**, allowing easy interfacing with legacy protocols or PLC Programming through standard industrial interfaces. Furthermore, the device features **built-in data visualization capabilities**, enabling users to monitor key parameters and process trends directly from the controller. The integration of these advanced features makes the HX controller a powerful and versatile edge solution that complements the Unified Gateway in achieving seamless, intelligent, and scalable industrial data management.

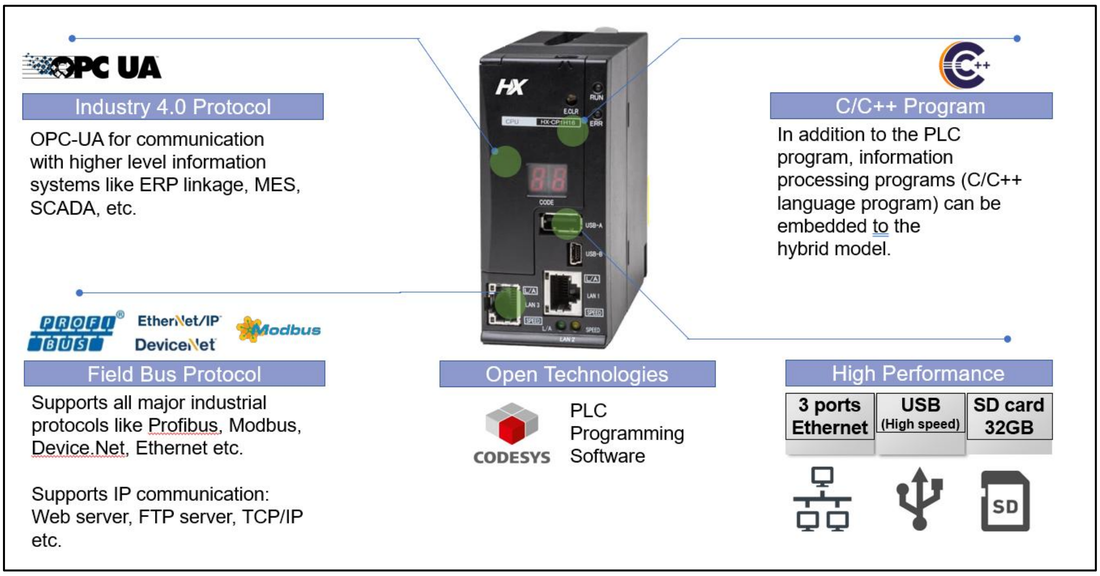


Fig: HX-Hitachi Controller Overview

4.2.4 Real-Time Log Monitoring:

This solution includes a **built-in real-time log monitoring** functionality that allows users to instantly view system status, communication logs, errors, alarms, and events from a respective screens or dashboard. This eliminates the need to manually search through large log files and helps engineers quickly identify and resolve communication or configuration issues.

The live monitoring capability provides complete visibility of data flow and connection health across all active protocols. It ensures faster troubleshooting, reduces downtime, and maintains transparency in data exchange between plant systems and the centralized server.

4.2.5 User Operation Tracking:

The **Unified Gateway Solution (UGS)** maintains a comprehensive **user operation tracking** feature that records every action performed within the system. This includes configuration changes, user logins, protocol updates, and system operations, creating a complete audit trail for accountability and traceability. By logging into all user activities, UGS enhances system transparency and supports secure, role-based access management. This ensures that all configuration and operational actions are properly documented, helping maintain compliance with plant IT and data governance standards.

4.2.6 Multiple Protocol Support

The **Unified Gateway Solution (UGS)** is designed to support a wide range of **industrial and IoT communication protocols**, enabling seamless data exchange between diverse automation systems and enterprise applications. It natively supports **Modbus, OPC DA, OPC UA, UDP, TCP/IP, MQTT**, and **DBlink (SQL Server, Oracle, MySQL)**, providing flexibility to integrate both legacy and modern systems within a unified framework.

This multi-protocol capability allows the UGS to act as a central communication hub across all plant units, reducing dependency on multiple gateway tools. Its modular architecture ensures scalability, enabling new protocol integration in the future without disrupting existing configurations or operations.

4.2.7 Advance House Keeping Features for Space Utilization

The **UGS** also offers an advanced housekeeping feature designed to optimize database and storage space utilization. This functionality automatically monitors data growth and manages historical records by applying configurable data retention and cleanup policies. Old or redundant data is archived or purged based on defined time intervals, ensuring that system performance remains consistent over long-term operation. This proactive housekeeping mechanism minimizes manual maintenance effort, maintains optimal database health, and ensures efficient space utilization within the data concentrator environment.

4.2.8 Email Alert and Reporting System(Value-Added):

This feature will send data acquisition reports, log data, etc. to specified persons at a predefined frequency and timing. Detailed contents on this reporting feature will be finalized during engineering discussion and mutually agreed.

Automating reporting system can deliver substantial **benefits** to customer. Reporting system automation will reduce the time spent reporting and allow your team to do more process optimization and revenue generating activities. **Reporting and email alert system** ensures that you can deliver more comprehensive and customized reports to key executives, more quickly.

Email reporting feature is provided with UGS System as a complimentary feature.

“Essential network configurations, tasks involved in connecting to the network, monitoring the network connection (including connection changes), giving users control over an app's network usage and authentications are to be decided and made/managed by buyer.”

4.2.9 Remote System Support (Value-Added):

SELLER will provide complimentary remote maintenance support for troubleshooting and issue resolution (if any) from the date of completion of project for first six (6) month period. Necessary infrastructure and network configuration will have to be enabled by BUYER to facilitate this. Remote maintenance support will be limited to scope of supply of this proposal. Remote support does not include additional features, modifications, and requests; its scope is limited to data gathering, system fine tuning, troubleshooting and issue resolution. Additional requests can be handled separately and not part of this support.

The effective working hours at HIL Office shall be from 9:00 AM to 5:00 PM IST (Monday to Friday), excluding National Holidays.

NON-DISCLOSURE AGREEMENT (NDA) will be signed between Buyer and Seller before starting the remote support.

4.2.10 User Accounts (Value-Added):

All the users of the Unified Gateway Solution are grouped into specific roles. Each role is provided with specific user rights (rights to access specific features). Any users with Administrator role can create/modify/delete roles and users. UGS can only be accessed by secure login credentials which enable unauthorized access security and to prevent system from any unwanted data flow failure due to human interventions.

4.2.11 Documentation Control

The **Unified Gateway Solution (UGS)** further includes a built-in **documentation and help system** with both **standard and search-based support features**, enabling users to quickly access relevant information for configuration, troubleshooting, or operational guidance. This integrated help resource minimizes downtime and assists plant engineers in resolving issues efficiently without dependency on external manuals.

The **Seller** shall also provide complete and comprehensive documentation covering all aspects of the system, including **general functional specifications, hardware and software specifications, system functions, wiring diagrams, utility documents, and interface documentation**.

4.2.12 Customer Training

SELLER will provide hands-on training on usage of system from operation perspective during commissioning to maximum of 5 persons for maximum of 3 days. Training shall cover mutually agreed topics from the scope of supply.

5. FUTURE SCALABILITY BENEFITS OF SOLUTION

NOTE: This section (Section-5) outlines the scalability features of the UGS. While these features are not included within the current scope of work, they may be contracted separately if required.

The **Unified Gateway Solution (UGS)** is designed with a modular and extensible architecture that allows seamless scalability for future integration with new process lines and/or utility system or energy system. Additional protocols, data sources, or production units can be integrated without major system redesign, ensuring long-term adaptability.

Some of the future scalability benefits of our solutions are as below :

- **Future Scalability & Integration**:

  The UGS architecture supports **future scalability for integrating additional production lines or machines**. It can be easily extended to communicate with new **PLCs or control systems** using standard OT protocols, creating unified plant-wide monitoring and controlling environment. This ensures that future expansions—such as new rolling mills, process units, or auxiliary systems—can be incorporated within the same Unified Gateway infrastructure without redesigning the entire communication framework.

  The **Unified Gateway Solution** is built on an open and flexible communication framework that supports **interoperability with diverse plant and enterprise systems**. It can seamlessly integrate with **Manufacturing Execution Systems (MES)** and **Enterprise Resource Planning (ERP)** platforms, automating data exchange between production and management layers. This integration enhances decision-making by providing accurate, real-time production data to planning and resource management systems, thus enabling synchronized operations and improved production efficiency.

  The system offers future feasibility for seamless **on-premises, hybrid, or off-premises cloud integration,** enabling secure data storage, remote monitoring, and real-time access from any location.

- **Data Acquisition – High-Frequency Intelligent Data Collection :**

  The **Unified Gateway Solution (UGS)** framework enables data collection at very high frequency like 10msec or 50msec. The UGS system can be used for high speed data acquisition of **intelligent process data** (like 10% of overall tags**)** and it can be analyzed using Hitachi **AutoML platform** on a local on-premise server.

  This setup allows predictive and anomaly detection models to run at the edge, delivering actionable insights in real time while ensuring scalability and intelligence across the system. The integrated Hitachi **AutoML Platform** provides an automated analytics environment that handles **data preprocessing, feature engineering, model selection, evaluation, and hyperparameter tuning** without manual intervention. Through these capabilities, the combined UGS and AutoML framework generates **predictive and prescriptive insights** that help optimize production performance, enhance equipment reliability, and improve overall efficiency — establishing a robust foundation for AI-driven operational excellence and digital transformation within the plant.

  In addition, the UGS is engineered for large-scale and **high-speed data acquisition**, It supports critical tag collection at frequencies **as low as 10 milliseconds** ensuring high-resolution data capture for rapid process.

- **HX Gateway Capability – Ability to Send Control Outputs to PLCs :**

  The HX-Hybrid Hitachi Gateway Device extends beyond data acquisition by offering advanced control capabilities. It supports **sending control outputs directly to PLCs**, enabling the execution of automated commands such as **setpoint adjustments, control loops, or logic triggers** from supervisory or edge systems. This transforms the HX Gateway into a dual-purpose device capable of both data concentration and decentralized control.

  In addition, the HX Gateway supports **PLC programming in multiple IEC 61131-3 standard languages**, including Ladder Diagram **(LD)**, Structured Text **(ST),** Function Block Diagram **(FBD),** and Sequential Function Chart **(SFC).** This flexibility allows engineers to create, modify, and deploy control logic directly on the device for localized automation. Such programmability enables the future implementation of edge-level control applications, remote automation strategies, and quick system adaptations without dependency on central PLC reprogramming.

- **SMS Alert system**:

  It is important to take actions/decisions quickly to have the desired effect and reduce unwanted productivity or quality loss. For this purpose, SMS alert system provides real-time updates to plant executives about significant events in the processing line. Contents of SMS and which events should trigger SMS alerts can be customized in this system.

- **Edge Computing Configuration:**

  The solution can be extended to supports **edge computing**, enabling data processing closer to the source to reduce latency and network load while improving real-time responsiveness. HX Gateway devices can perform **local computations**, condition monitoring, and data filtering before sending optimized information to the central concentrator. Future configurations will include rule-based logic and data transformation at the edge to enhance data quality and relevance, making the Unified Gateway a scalable and future-ready industrial platform.

6. TECHNOLOGY STACK

The technology stack for various components is as follows:

6.1 BASIC SOFTWARE SPECIFICATIONS

The UGS system will be offered under ".Net Framework" platform. SELLER shall consider using latest version before detail design start. The software specifications for ".Net Framework" environment is as under:

Components	Technology Used/Make	Q'ty
Application Front-end	Material Bootstrap, TypeScript, HTML5, CSS3. Application can be viewed on a standard web browser like Chrome, Edge & Mozilla	-
Database	MS SQL Server 2022 Standard Edition	1
Application Back-end	C#, .NET Core, ASP.NET API, SignalR	-
Operating System	Windows Server 2022 Standard Edition OEM with 5 User CAL	1
Framework	.Net Framework 8	-
Development	Visual Studio Code, SSMS, Codesys	-
Antivirus	Trend Micro for Server	1

6.2 BASIC HARDWARE SPECIFICATIONS

SELLER shall consider using latest version before detail design start. SELLER will submit documents to be approved by the buyer. IP address, domain for the network shall be decided mutually by Seller & Buyer.

No.	Equipment name	Maker / Type	Quantity	Specification
			Main	
1	UGS Server (Data Concentrator)	HP/Dell/Lenovo Server (Tower based) Or equivalent.	1	Intel Xeon-Gold Processor 32 GB DDR4 RAM, 2 TB Memory 4 Ethernet LAN ports (1Gbps), USB Keyboard and mouse
2	UGS Server Console	HP/Dell/Lenovo Monitor	1	FHD (1920 x 1080) resolution monitor – 24inch with VGA and/or HDMI
3	Ethernet Switch (managed)	HIL Approved	1	
4	Ethernet Cable (200m length)	Any reputed make	1 Set	
5	HX Controller & Junction Box	Hitachi	1 Set	For UGS (In case of Option-1)
6	HX Controller & Junction Box	Hitachi	4 Set	For UGS (In case of Option-2)

6.3 THIRD PARTY SOFTWARE REQUIREMENTS

Remote Link: To provide a suitable level of response to operation & process execution problems and queries raised on site, Seller requires a network connection via broadband / VPN to be established between Seller and the Buyer's site. Set-up and maintenance of this link will always be under the BUYER's authority.

6.4 SYSTEM ARCHITECTURE

Overview of the Unified Gateway Solution reference architecture is shown below:

1 HIL is providing Two Option for Architecture, based on Customer choice, HIL will submit the proposal.

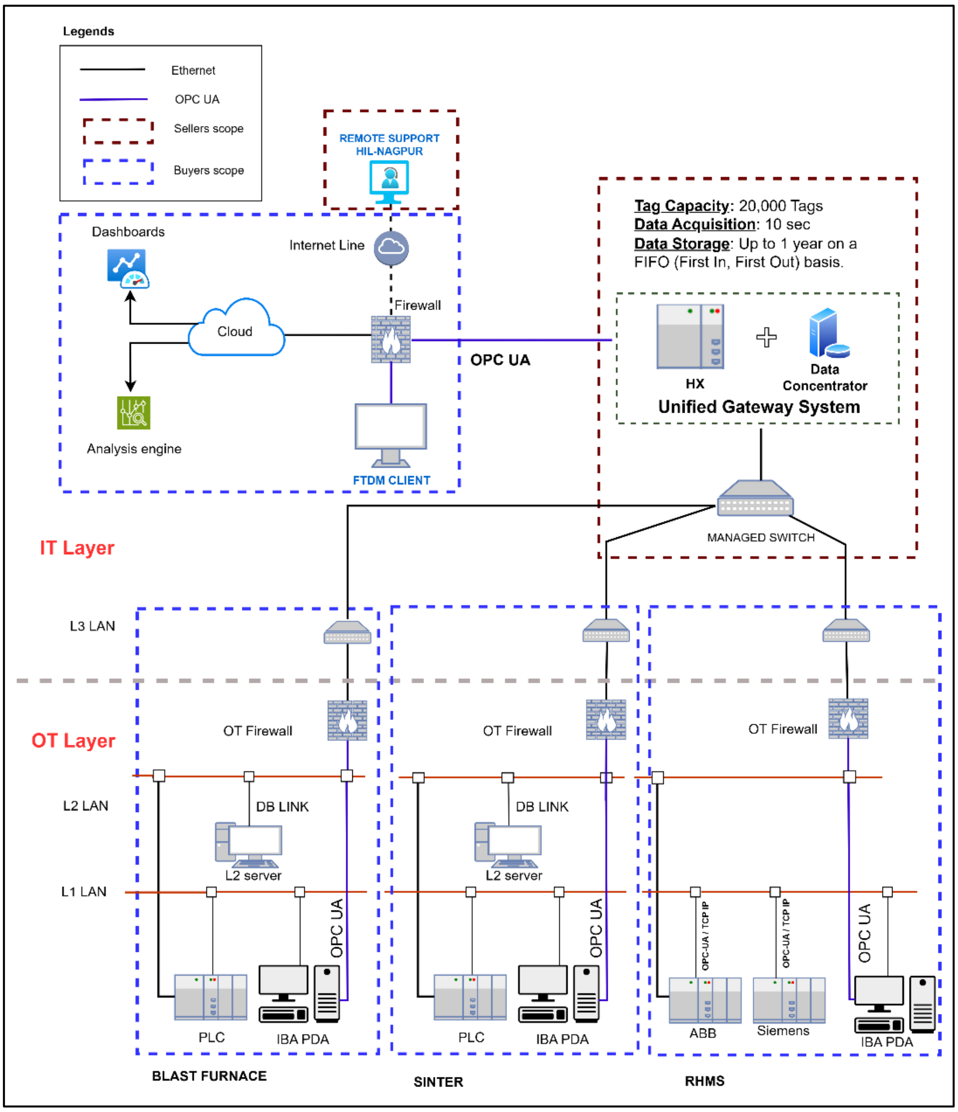


Fig: System Configuration (For reference only) (Option-1)

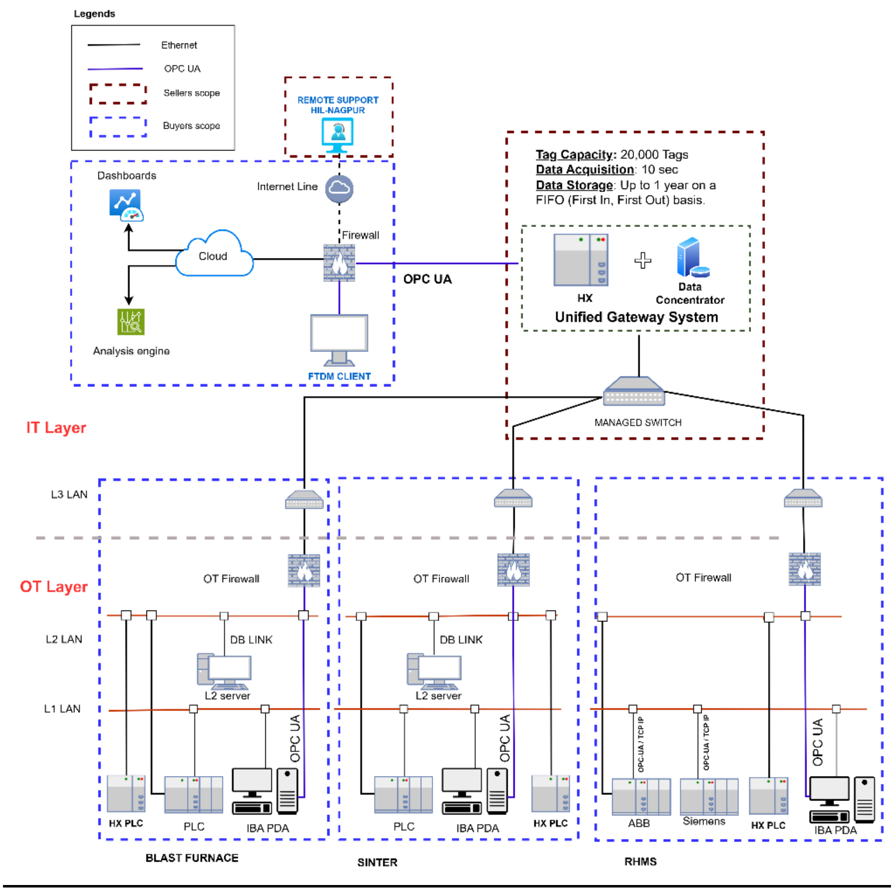


Fig: System Configuration (For reference only)(Option-2)

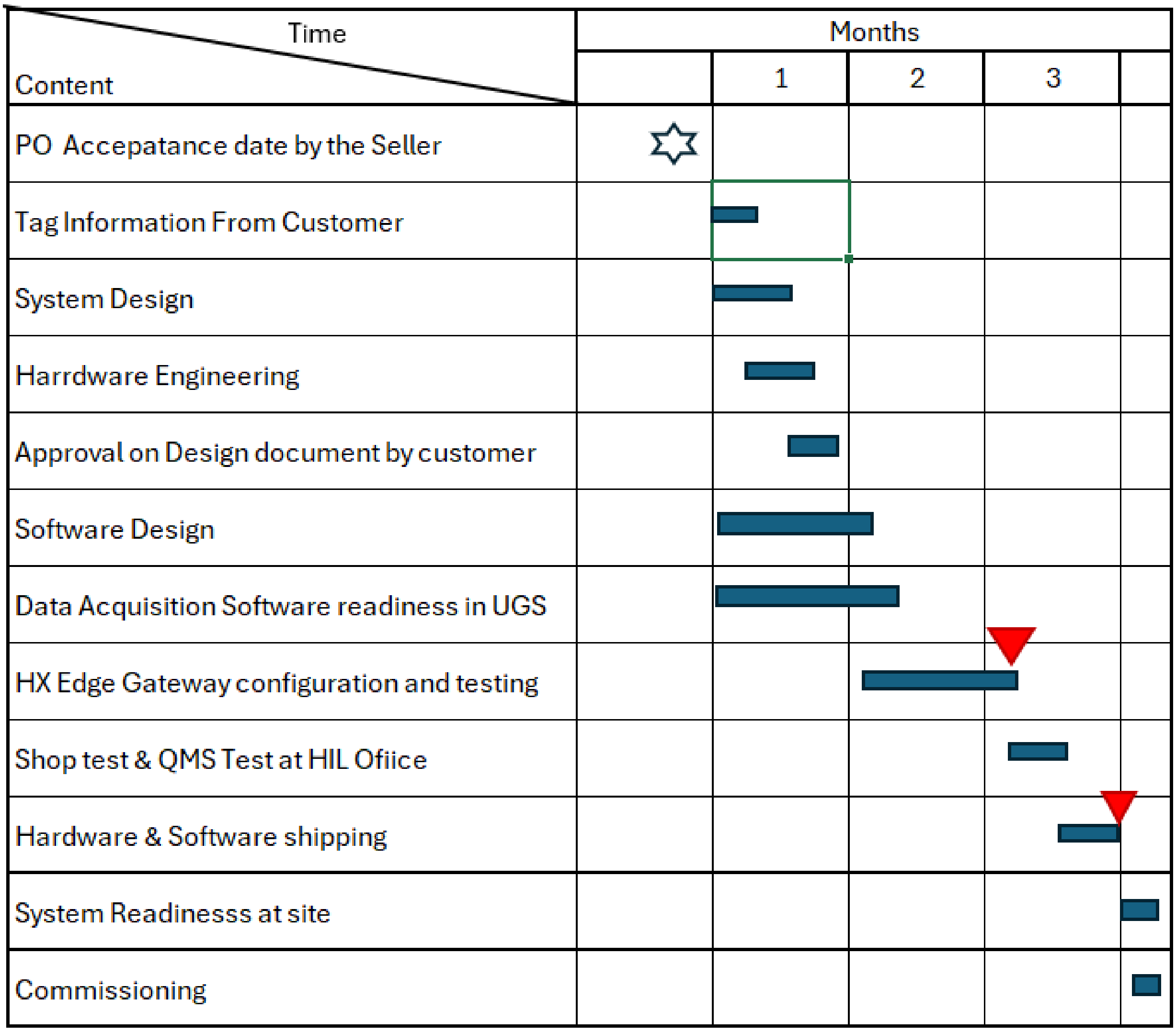
Note for kind information for UGS:-

Unlimited Tag Capacity: Practically no limit on the number of tags that can be handled.

High-Speed Data Acquisition: Supports data sampling rates as low as 10 milliseconds.

Long-Term Data Storage: Stores data for up to 1 year on a FIFO (First In, First Out) basis.

6.5 OVERALL SCHEDULE INDICATIVE FOR REFERENCE ONLY



Note: After PO Approval on Seller will take 3 Months for Software Configuration & Testing. In the event of a delay in System design document approvals from the Customer, it will lead to an overall delay in the delivery. Above delivery schedule is for UGS application for reference only.

The number of day's site-supervisor deputed to the site for the commissioning, deployment & training at site:

- Total 10 Man-days (Inclusive of on-site training).
- Travelling days are not included in the above days.

7. SCOPE OF SUPPLY

7.1 SCOPE OF SUPPLY DEFINITIONS

The engineering and software development service necessary for the proposed PDCMS system shall be provided either by Buyer or by the Seller in accordance with the Scope of supply List. The matters not described in the said list shall be determined through mutual discussion.

Buyer : JSOL, Angul (B: Buyer)
Seller : Hitachi India Pvt Ltd (S: Seller)

BD : Basic Data
BE : Basic Engineering
DD : Detail Design
SU : Supply or Service
ER : Erection
COM : Commissioning

X/Y: Primary Work/Services provided by X with Support from

7.2 DIVISION OF SUPPLY, ENGINEERING AND SERVICES

No.	ITEM	Responsibility					
		BD	BE	DD	SU	ER	COM
(1)	**Services**						
-1	Project Execution	B/S	B/S	B/S	-	-	-
-2	Overall system design	S	S/B	S/B	-	-	-
-3	Work Test and Simulation	S	S	S	-	-	-
(2)	**System Engineering**						
-1	UGS Hardware design	S	S	S	-	-	-
-2	UGS Network Design	B/S	B/S	B/S	-	-	-
(3)	**Hardware**						
-1	Managed Switch (1No.)	S	B/S	B	S	B	B/S
-2	Rack for Network Switches / HUB. With Power Supply Extension	B	B	B	S	B	B/S
-3	Ethernet cables	S	S	S	S	B	B/S
-4	UGS Server	S	S	S	S	B	S
-5	HX Gateway Device	S	S	S	S	B	S
-6	HIL approved Panel Box (for HX)	S	S	S	S	B	B/S
-7	Desk for Server along with Power Supply	B	B/S	B	B	B	B
-8	Firewall	B	B/S	B	B	B	B

-9	GSM Gateway (for SMS)	B	B/S	B	B	B	B
-10	SIM for GSM Gateway	B	B/S	B	B	B	B
(4)	**Software**						
-1	Section 6.1	S	S	S	S	-	S
(5)	**Interface with External Systems**						
-1	Interface with IBA PDA system (at BL)	B/S	B	B	B	-	B
-2	Interface with IBA PDA system (at SIN)	B/S	B	B	B	-	B
-3	Interface with IBA PDA system (at CON)	B/S	B	B	B	-	B
-4	Interface with IBA PDA or PLC system (at RMHS side)	B/S	B	B	B	-	B
-5	Interface with IBA PDA system (at UGS side)	B/S	S	S	S	-	S
-6	Interface with PLC system of RHMS (at PLC side)	B/S	B	B	B	-	B
-7	Interface with PLC system of RHMS (at UGS side)	B/S	S	S	S	-	S
-8	Interface with L2 system of Sinter & BF (at L2 side)	B/S	B	B	B	-	B
-9	Interface with L2 system of Sinter & BF (at UGS side)	B/S	S	S	S	-	S
-10	Email Server SMTP (for email reports)	B	B/S	B/S	B/S	-	B/S
(6)	**TRAINING**						
-1	Training (Max 3 days) for 5 persons	S	S	S	S	-	-
(7)	**CONSTRUCTION & ENGINEERING**						
-1	**Cables (Wherever Required)**						
	Power cables	S	B	B	B	B	-
-2	**Erection Materials**						
	Power and control junction boxes	B/S	B	B	B	B	-
	Cable Trays, conducts, lugs, Glands & other cabling material	B/S	B	B	B	B	-
-3	**Earthing System**	B/S	B/S	B	B	B	-
(8)	**Pre-Engineering Activities**						
-1	Study of Existing system	B/S	B/S	B/S	-	-	-
-2	Design of L1(IBA PDA) – UGS interface	B/S	B/S	B/S	-	-	-

(9)	**Documents to be submitted for Reference & Records**						
-1	System Operation Document	S	S	S	S	-	-
-2	Hardware Specification	S	S	S	S	-	-
-3	System Interface Document	S	S	S	S	-	-

Note: Any additional requirements beyond the scope mentioned above shall be discussed and mutually agreed upon. A separate proposal will be submitted for such additional requirements.

7.3 WORK COMPLETION CERTIFICATE

Upon the Completion of the work (based on the below criteria), the Work Completion Certificate will be issued by the Buyer to the Seller, signifying the satisfactory completion of the project.

Work Completion Criteria:

- Supply of Hardware & Software as per the scope of supply (described in section 6.1 & 6.2)
- Completion of Commissioning works Man-days (as described in section 6.5)
- Submission of all documentation as per the scope (described in section 7.2 (9))

In case if the Work Completion Certificate is not issued by BUYER within this period, it is considered that Work Completion Certificate as deemed issued by the BUYER. If there are any issues/problems faced by BUYER, it will be mutually discussed, and SELLER would support BUYER with the best of the ability to resolve the issue. By issuing the Work Completion Certificate, the Buyer acknowledges that the Seller has fulfilled all contractual obligations and requirements.

7.4 BUYER OBLIGATIONS

The Buyer should fulfil the following obligations

- Responsible for the project execution (answer technical queries in reasonable time and coordinate with all the stake holders of the project)
- Arrange all the hardware in Buyer scope well ahead of the agreed time schedule.
- Network cables & accessories not included in this proposal to be provided by Buyer
- Site access to Seller's representative for data collection and discussion with the technical team as per requirement.
- Dedicated internet connection
- In case of any health issue to Seller's representative, buyer to immediately provide best available medical facility. The expenses will be borne by the Seller.
- Latest Level-1, IBA PDA server Software Details and Tag Details along with any existing HMI DB Schema (if exist) to be provided to Seller
- Configuration at Level-1 IBA PDA server across plant for interface will be done by Buyer only.
- If required, availability of 3rd party equipment engineers at site for engineering discussions and commissioning.
- SMTP details for automated email alerts.
- Essential network configurations, tasks involved in connecting to the network or internet, monitoring and configuration of the network for email alert will be done by buyer.

7.5 EXCLUSION LIST

Only the services mentioned in the Scope of this proposal are included. Any further services need to be mutually discussed, and commercial / technical implications discussed and compensated through a previously agreed change management mechanism.

Specific attention is drawn to the following services that are excluded from the scope of this proposal (unless specifically included elsewhere in this proposal).

In this proposal following are not included:

- Interface with external devices other than explicitly described in the proposal document.
- Software patches including but not limited to Microsoft Security updates, Antivirus updates or any other software upgrades or patches.
- Any warranty, guarantee, liability, responsibility, etc. about productivity, quality, yield, etc.
- Any equipment, components, tools & works not described in this proposal.
- Source code of software of core technology.
- Modifications to existing electrical and automation systems including but not limited to existing Level-1, Level-2, MES, SYC, CYC system.
- Erection activities wherever required for project execution. Any kind of Civil work.
- Hardware and software other than that is mentioned in technical document.
- Mechanical equipment supply, modification etc.
- Support for troubleshooting for mechanical/operation/process issues of customers.
- PDA software and related hardware
- Firewall and other networking components
- Performance with respect to productivity, yield, quality, process capability, process performance, etc.

7.6 BINDING CONDITIONS

- SELLER scope of supply & services as described in agreed & signed technical specifications (TS) shall be limited, as per the inputs/information obtained from the BUYER, only to the modification/creation of interface program for helping the BUYER to use SELLER supplied equipment/items for obtaining the required functioning at BUYER own responsibility.
- All the services, except for those under the scope of SELLER as per the agreed & signed TS, will be under the full responsibility of BUYER. SELLER neither guarantees any performance nor SELLER is liable for any claims for the non-performance of such Equipment & services.
- SELLER scope of supply & services excludes any supervision, management, regulation, arbitration and/or measurement of BUYER's personnel, agents or contractors and work related thereto nor does include any responsibility for the scheduling, monitoring or management of their work.
- SELLER supervisors shall have the right to leave the site without any permission or approval of BUYER in the following cases:

a. If the idling periods extend more than a considerable time as judged by SELLER.
b. If the BUYER does not follow the conditions of the agreed terms as per the agreed and signed TS
c. An act of God such as natural calamities, war, riot, rebellion, invasion, commotion, epidemics, terrorism, or such other phenomenon which is predictable to affect the body or life of our supervisors directly or indirectly.
d. If BUYER prerequisites are not fulfilled as per the agreed and signed TS.
e. Circumstances at project site are not conducive and are beyond SELLER's control as judged by SELLER.

- SELLER will not be responsible for

  a. Performance guarantee of any kind.
  b. Productivity AND/OR Quality of the Product

- SELLER will not be responsible for any failed/missing/damaged equipment, item, and components etc. during the execution of the service work unless otherwise explicitly mentioned in the contract.
- Unless explicitly mentioned in the agreed and signed TS, SELLER will not take responsibility for any modification or updating of the existing system.
- The effective date of execution of the contract shall be the PO acceptance date by the SELLER.

7.7 CYBERSECURITY DISCLAIMER

SELLER strongly recommends that security updates are implemented by Buyer as soon as they are available and that the latest versions are used. Use of versions that are no longer supported, and failure to apply the latest updates may increase the exposure to cyber threats. This application contains confidential information. It is the sole responsibility of end-user to maintain the confidentiality and proper usage of the information contained herein. The BUYER is responsible for preventing unauthorized access to the existing plant systems. The BUYER will be responsible for Cybersecurity, Information Security, Hardware Security, Personal Data Security, Overall Data security, Network security, equipment security. In no event shall SELLER be liable for any direct, indirect, incidental, consequential, or special damages arising out of or in connection with any cybersecurity incidents, data breaches, unauthorized access, or any other security-related events, regardless of the cause of action.

8. DISCLAIMER

8.1 SOFTWARE LICENSES

The use of the delivered software, or parts of the same, is strictly limited to the services provided. The buyer has the right to use and modify the software for normal use and for the lifetime of the system supplied by Seller. Any duplication (except for normal back-up purpose), distribution to third parties, sell or use in other plants is prohibited. It's also prohibited to make software accessible to third parties. All rights of the software remain with Seller.
'BUYER is responsible for maintaining and ensuring valid license for use of third-party software and legal use of third-party software used in this system. In no event shall Seller be liable for any issue or claim or damage arising out of use of any third-party software.'

8.2 CHANGES DUE TO TECHNICAL IMPROVEMENTS

Seller follows a policy of continuous improvement of design, manufacturing and software development and therefore reserve the right to supply systems differing in detail from what is described in this document.

8.3 CONFIDENTIALITY OF INFORMATION (Non-Disclosure Agreement)

According to the law, Seller considered this document to be a company secret and therefore prohibits any person to reproduce or disclose it in whole or part to third parties, without specific written authorization of Seller Management.

8.4 LIMITATION OF LIABILITY & CONSEQUENTIAL DAMAGES

With respect to any and all liability of HIL out of or in connection with this proposal notwithstanding anything contained herein to the contrary in no event shall HIL's total liability for any and all claims, damages or liabilities arising out of related to its performance of this proposal, including but not limited to any liquidated damages, costs for repairs, replacement, rectifications of defects, dismantling, etc., exceed one hundred percent (100%) of the HIL's Contract Price. In no event shall HIL be liable for any indirect or consequential damages, including but not limited to any loss of profit, loss of use, loss of Contracts, loss of production or business interruption. According to law, SELLER considers this document to be a company secret and therefore prohibits any person from reproducing or disclosing it in whole or part to third parties, without specific written authorization of SELLER Management.

9. FREE PROOF-OF-CONCEPTS

Seller is pleased to provide a **complimentary Proof of Concept (PoC)** along with the aforementioned system to showcase SELLER's latest offerings in Composite Automation & Digital Solutions (CADS) & Digital Transformation (DX) Solution. The PoC will be conducted on the same server.

The solutions under the PoC will be deployed for a limited period of 2-months to demonstrate the capabilities of these solutions. Upon successful completion of the demonstration, the BUYER must capitalize on the benefits by purchasing these solutions through a separate commercial contract, ensuring continued access to their full potential. The SELLER reserves the right to withdraw the PoC solutions if the BUYER does not proceed with the purchase following the demonstration.

9.1 AUTOML PLATFORM

To drive Digital Transformation (DX) in the existing setup, Seller is pleased to propose the deployment of an "AutoML Platform" on the existing LPAS server for a limited period as a Proof of Concept (PoC).

The objective of this engagement is to demonstrate the platform's ability to accelerate data-driven insights, streamline model development, and enhance decision-making capabilities without extensive manual intervention.

- The AutoML platform will be installed and configured on the existing UGS server infrastructure.
- The PoC will run for a limited duration solely for demonstration purposes.

The platform will enable: Automated data preprocessing, feature engineering, and model selection.

Model evaluation, and performance benchmarking.

Hyperparameter Tuning

Demonstration of ML Model working through predictive and prescriptive insights.

Seller scope will be limited to the following items for the PoC on AutoML platform:

- Supply for AutoML system software in the existing server.
- Assist Buyer in integrating UGS data with the AutoML platform for usage
- Assist Buyer in developing atleast 2 supervised learning ML model using the available data for demonstrating the usage of the platform.
- The data size considered for the development of supervised ML model will be limited to 6-month data.
- Only specific features of the AutoML platform will be supplied for demonstrating the capabilities of the system. Advanced capabilities, integrations, or customizations are excluded from this PoC.
- The PoC is intended solely for evaluating the feasibility and performance of the AutoML platform in a controlled environment. It does not constitute a commitment to full-scale deployment or commercial use.
- The PoC is provided "as-is" without any warranties, service-level commitments, or ongoing support obligations.

- All intellectual property related to the AutoML platform remains the sole property of the seller. No rights are transferred or implied through the PoC.

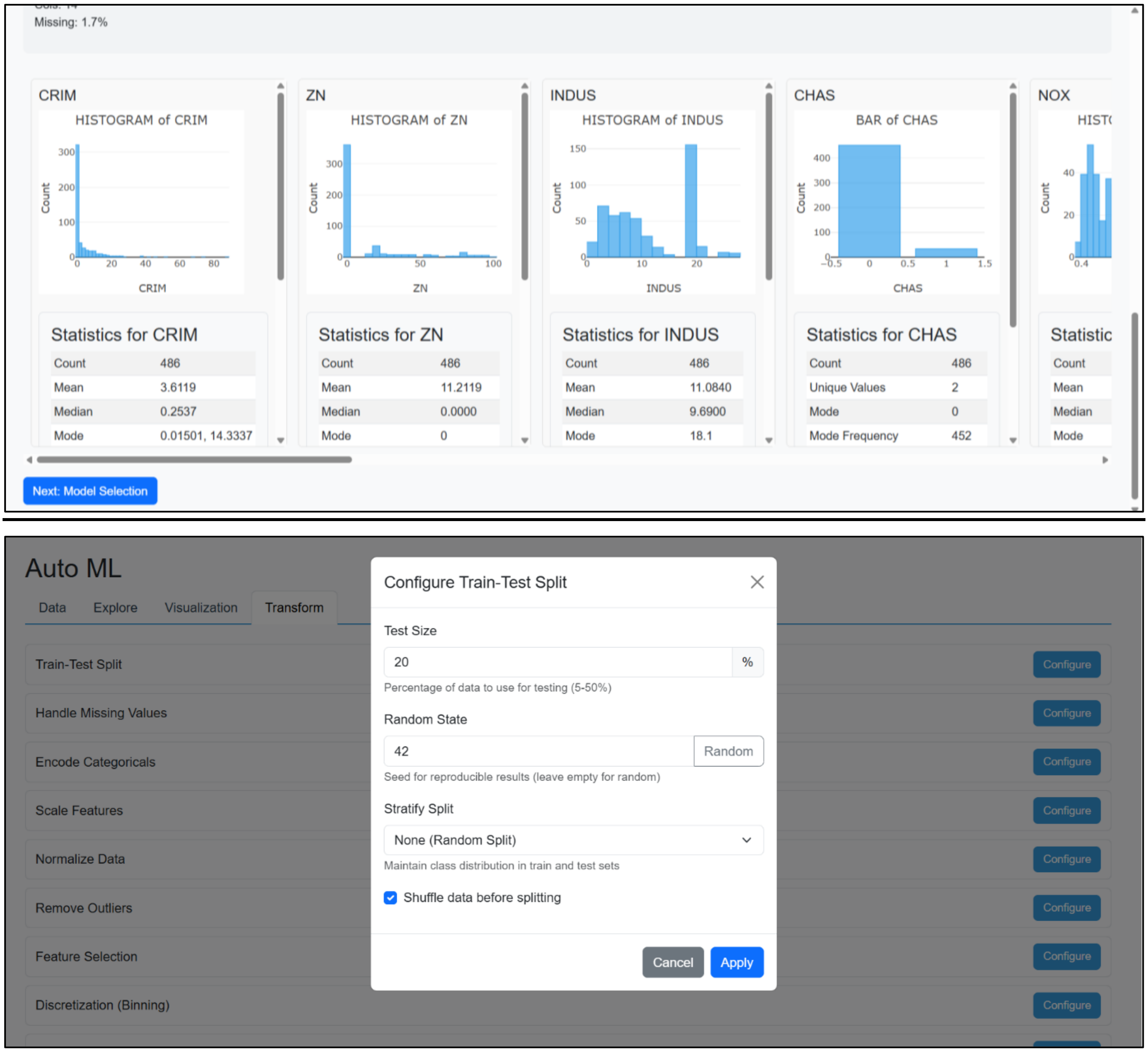


Fig: AutoML Platform Reference Screenshot

End of Technical Proposal